

NGUYEN VO XUAN DAI

AI Software Engineer

A dedicated AI Software Engineer with over 4 years of experience specializing in the design and development of robust, scalable intelligent systems. Expert in bridging the gap between deep learning research and high-performance industrial applications, with a versatile background spanning both Generative AI and Computer Vision domains. Proficient in architecting end-to-end AI pipelines, optimizing large-scale model deployment for production environments, and engineering complex data processing systems to unlock enterprise insights. A results-driven professional with full professional English proficiency, committed to delivering tangible business value through innovative AI solutions and technical leadership in global, collaborative environments.



Contact

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Education

DANANG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Major: Control Engineering & Automation

Technology Degree: Engineer

Classification: Good

Certificates

★ TOEIC Certificate (975/990)

Work Experience

AI Engineer (Team Leader) – THACO INDUSTRIES's R&D CENTER

06/2022 - 06/2024

- **Spearheaded** and delivered AI-driven Digital Transformation (DX) initiatives, **driving 5x faster parking operations** and **streamlined multi-user attendance tracking**.
- **Owned** the end-to-end AI/ML lifecycle, including designing, training, and optimizing DL models, and production deployment via robust **MLOps** pipelines.
- **Leveraged** expertise in **Linux OS/Embedded Systems** to deploy **integrated IoT/AI applications** on **NVIDIA Jetson** platforms for specific use cases.
- **Managed** a team of **4 engineers**, providing mentorship and conducting regular code reviews, **driving successful completion of key technical milestones**.
- **Developed** and **published** comprehensive technical documentation and internal training materials.
- **Coordinated** and **delivered** internal technical training sessions and specialized technology workshops.

AI Software Engineer – Techvify Software Vietnam (Danang)

06/2024 - 12/2025

- **Secured international client projects** by presenting high-impact technical solutions and detailed proposals, **significantly expanding** the company's AI service portfolio.
- **Engineered** and **maintained robust backend services** and complex **Batch/Cron jobs** using Java and Bash scripting, guaranteeing **99.9% uptime** for critical data processing tasks.
- **Specialized in embedded systems optimization**, fine-tuning and deploying high-performance AI applications on **NVIDIA Jetson Edge devices** to achieve **sub-50ms inference latency** for client solutions.
- **Contributed to consistent project success** by utilizing the **Agile Scrum methodology** and implementing **CI/CD automation**, ensuring milestones were **delivered reliably** and on schedule.
- **Enforced rigorous code quality standards** through comprehensive testing and peer reviews, **significantly enhancing system stability and reliability** in the production environment.
- **Spearheaded knowledge transfer initiatives** by designing and conducting technical workshops on specialized cutting-edge topics including **Generative AI/LLM, AI Agents**, and **AI-powered Tools** for the development team.

AI Engineer – Saigon Technology (Danang)

12/2025 - Present

- **Architected and delivered an Enterprise Unified AI Search ecosystem**, enabling seamless natural language querying across multi-modal datasets and complex relational databases to accelerate internal knowledge discovery.
- **Optimized model serving infrastructure and deployment strategies**, significantly increasing system throughput while reducing hardware resource overhead for high-concurrency production environments.
- **Engineered an automated intelligent data pipeline** that transformed fragmented, unstructured corporate documents into high-integrity searchable assets, ensuring data consistency and precision.
- **Developed and scaled a high-precision Computer Vision monitoring system**, automating industrial inventory tracking and achieving an 85% reduction in manual auditing time for manufacturing facilities.
- **Established a robust MLOps lifecycle and observability framework**, streamlining the transition from experimental models to production-ready solutions with continuous performance tracking and cost-effective scaling.

Technical Skills

Programming Languages	Experienced	<i>Python</i>
	Familiar With	<i>Java, JavaScript</i>
AI Development	Generative AI & LLM Ops	<i>LangChain, LangGraph, vLLM/Ollama, GGUF/AWQ Quantization, Text2SQL, Langfuse</i>
	Data Ingestion	<i>Docling, TesseractOCR, PaddleOCR</i>
	Other Frameworks	<i>PyTorch, Pandas, OpenCV, Ultralytics YOLO, MLflow, NVIDIA TensorRT, RASA, Gradio</i>
Backend	Frameworks	<i>Django REST Framework, FastAPI</i>
	Script-Based Automation	<i>Bash/Shell scripting, Batch jobs / Cron jobs</i>
DevOps	Deployment Tools	<i>Docker, Docker Compose, Jenkins</i>
	Infrastructure	<i>Windows Server, NVIDIA Jetson, AWS EC2</i>
Database	SQL	<i>PostgreSQL, Oracle, MS SQL Server</i>
	VectorDB	<i>Redis, Qdrant</i>
	GraphDB	<i>Neo4j</i>
Work Platforms	Task Management	<i>Jira, Trello, Clickup</i>
	Source Control	<i>GitHub, GitLab, Bitbucket</i>

Other Skills

Language Proficiency	English Communication (Proficient)	
Professional Skills	Prompt Engineering	Teamworking
	Technical Presentation	Research & Analysis
	Proactive Communication	Requirement Analysis
	Agile Methodology / Scrum Framework	Time Management
	Continuous Learning / Self-learning	

Highlight Projects

1. Enterprise AI Search & Knowledge Discovery

Role	AI Engineer
Project Description	An enterprise-grade search platform allowing multi-modal querying across structured relational databases and massive unstructured datasets (PDFs, Office docs, Images). The system integrates advanced language models to provide a natural language interface for complex data retrieval, ensuring high accuracy and low latency for internal knowledge discovery.
Responsibilities	- Designed the end-to-end system architecture, integrating Text2SQL reasoning agents with vector-based semantic search for a unified user experience. - Developed the intelligent data ingestion pipeline utilizing advanced parsing libraries and OCR engines to extract structured information from messy, unstructured enterprise documents. - Engineered the model serving infrastructure using vLLM, including custom quantization workflows (AWQ) to optimize resource utilization on production GPUs. - Integrated observability tools to monitor model traces, audit prompt performance, and track token consumption for cost-effective scaling.
Achievements	- Enhanced system versatility and retrieval precision by architecting a hybrid retrieval engine that intelligently routes queries between Text2SQL for structured data and Semantic Search for unstructured documents. - Reduced VRAM footprint by 40% while maximizing throughput for high-concurrency environments through model quantization (AWQ) and optimized deployment via vLLM.

	● Boosted data integrity and retrieval accuracy by engineering a multi-stage ingestion pipeline featuring automated OCR, semantic parsing, and deduplication. ● Ensured production-grade reliability by implementing a rigorous evaluation framework utilizing SQL-based verification and Langfuse metrics to monitor RAG performance.
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2. AI-Powered Tool Inventory Monitoring System

Role	AI Engineer
Project Description	An intelligent computer vision solution for automated inventory tracking of industrial tool drawers. The system leverages instance segmentation and object detection models to identify missing or misplaced tools in real-time from captured images, ensuring operational safety and tool accountability across manufacturing facilities.
Responsibilities	● Architected end-to-end MLOps pipeline combining object detection, instance segmentation, and advanced transformation algorithms for accurate tool localization and identification. ● Engineered modular inference system with flexible model switching capabilities, supporting GPU acceleration via NVIDIA TensorRT and CUDA optimization. ● Implemented automated training pipeline with hyperparameter tuning, model evaluation metrics, and S3-based dataset management for scalable model deployment across multiple drawer configurations. ● Developed synthetic data generation framework leveraging augmentation techniques to overcome small dataset constraints, creating 3x more training samples and improving model generalization across diverse drawer configurations.
Achievements	● Achieved 95%+ accuracy in tool detection and segmentation by implementing a multi-model ensemble approach that ensures robust performance under challenging lighting conditions. ● Reduced manual tool auditing time by 85% through the deployment of an automated real-time inference pipeline and immediate visual feedback systems. ● Accelerated development cycles and model reliability by establishing a full-scale MLOps pipeline covering the entire lifecycle from annotation and experiment tracking to version control. ● Improved operational workflow efficiency by integrating real-time visual alerts, significantly reducing human error in equipment management.

3. Real-Time Industrial Rock Fragment Size Analysis System

Role	AI Software Engineer
Project Description	A real-time AI-based solution for analyzing rock fragment sizes from live video streams. The system automatically detects when material is being transported, segments individual rocks, and calculates their real-world dimensions to identify oversized fragments, providing immediate feedback, alerting operators of anomalies.
Responsibilities	• Developed the end-to-end pipeline combining Instance Segmentation models (to isolate individual rocks) with advanced perspective transformation/calibration algorithms for accurate size calculation. • Engineered the final alert mechanism by integrating the detection application with an adjacent ESP32 microcontroller via Bluetooth to trigger the physical indicator lights. • Established comprehensive MLOps logging to record operational data and save analysis/sample images for continuous model enhancement and auditing. • Streamlined remote deployment by writing automation BAT/Bash scripts, enabling non-technical operators to install and maintain the system. • Optimized the model inference on Edge Device (NVIDIA Jetson) to ensure high reliability and low latency (<100ms processing time) in a harsh industrial environment.
Achievements	• Achieved 90% accuracy in classifying rock fragment sizes and detecting oversized anomalies crucial for operational safety and equipment wear prevention. • Reduced downstream equipment damage and operational downtime by providing real-time alerts for oversized material.

4. Enterprise Self-Service NLP Chatbot Platform

Role	AI Software Engineer
Project Description	A high-availability, self-service platform enabling Marketing and Retail departments to independently access, create, and deploy custom NLP chatbots to enhance customer care and sales operations.

Responsibilities	• Architected the microservices-based platform, focusing on high availability, scalability (Horizontal/Vertical scaling), and multi-tenancy. • Developed the core backend APIs for chatbot creation, integration, and deployment using Django REST Framework. • Implemented the CI/CD pipeline (Jenkins + Argo CD + Kubernetes) and monitoring solutions (Logging) to ensure robust platform performance and rapid deployment cycles. • Designed the foundation for the NLP models, facilitating quick development of demo and production chatbots for business units.
Achievements	• Successfully deployed a High-Availability platform serving various business units/retail stores, achieving 99.9% uptime during peak operational hours. • Reduced time-to-market for new chatbot initiatives by 50% by enabling self-service creation and automated CI/CD deployment. • Demonstrated impact on sales by successfully deploying a chatbot prototype that handled 40% of initial customer inquiries, streamlining support staff workload.

5. Automatic-Parking System

Role	AI Engineer
Project Description	A real-time system that automatically identifies employee vehicle license plates (ALPR), controls the security fence, and provides simultaneous audio/visual notification of entry and exit events.
Responsibilities	• Designed the full system architecture, integrating RTSP video feeds with backend APIs for access control. • Developed and custom Deep Learning models for Object Detection (vehicle/plate localization) and Character Recognition (ALPR). • Optimized the DL model for low-latency inference on Edge/Embedded systems (NVIDIA Jetson).
Achievements	• Achieved 97.5% recognition accuracy on license plates under varying lighting and weather conditions. • Reduced employee entry/exit time by 5 times. • Implemented a low-latency inference engine (<150ms end-to-end latency) for instant barrier operation.

6. High-Accuracy Enterprise Facial Recognition Attendance System

Role	AI Software Engineer
Project Description	A high-accuracy system for automated employee timekeeping, capable of simultaneous, real-time facial recognition for multiple users without requiring direct camera focus. The system records entry/exit events and logs history for workforce management.
Responsibilities	● Architected the large-scale solution, focusing on integrating real-time RTSP streams with the timekeeping backend for seamless data logging. ● Implemented a robust Face Detection pipeline and deployed a state-of-the-art Face Vector Encoding Model for identification. ● Optimized the DL model and data structures to support simultaneous recognition of multiple faces at real-time speeds. ● Ensured data security and privacy compliance during integration with HR APIs and historical database storage.
Achievements	● Achieved 99% recognition accuracy for rapid and reliable employee identification across diverse environmental conditions. ● Processed and recognized multiple users simultaneously in real-time to prevent queuing at entry points. ● Reduced time-theft risks and administrative overhead by implementing the fully automated and high-throughput attendance logging system.